

PROJECT PROPOSAL: CHAPTERS 1 – 3

Project Title: **AgriShield**

Protecting Kenya's Food Security, Ensuring Future Sustainability.

A Hierarchical, Agentic AI Predictive Intelligence System for Crop Yield and Livestock Forage Risk in Kenya

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Agriculture is the backbone of Kenya's economy, contributing approximately 33% to the Gross Domestic Product (GDP) and employing over 75% of the rural population (World Bank, 2021). The sector is predominantly characterized by rain-fed smallholder crop farming and pastoralist livestock rearing. However, this reliance makes the sector highly vulnerable to climate variability, including delayed rains, prolonged dry spells, and sudden floods. According to the Food and Agriculture Organization (FAO, 2023), these climate shocks cause annual agricultural losses exceeding billions of shillings, threatening national food security. While meteorological data and satellite imagery (e.g., NDVI for pasture greenness) are increasingly available, the translation of this raw data into actionable, localized intelligence for county-level decision-makers remains a critical, unresolved challenge.

1.2 Problem Statement

Despite the availability of climate and agricultural data, Kenya's agricultural sector continues to suffer from unpredictable crop yield shocks and livestock forage deficits. Currently, Early Warning Systems (EWS) and county interventions operate **reactively**, deployed only after visible distress or crop failure has occurred. Furthermore, existing solutions are highly siloed, focusing either solely on crops or solely on livestock and output complex, technical GIS data that local agricultural officers cannot easily interpret or act upon (Hansen et al., 2019). Finally, traditional machine learning models suffer from a "black box" problem, lacking the interpretability and standardized reporting formats required for government budget allocation (Tzounas et al.,

2023).

Therefore, this project seeks to develop **AgriShield**, a hierarchical (county-to-region) predictive intelligence system. By integrating a champion machine learning model with an **Agentic AI Gria (via NVIDIA NIM)**, the system will autonomously translate complex predictive outputs into dynamic visualizations, plain-English explanations, and automated PDF reports, bridging the gap between raw data science and actionable agricultural policy.

1.3 Research Objectives

General Objective:

To develop an agentic, hierarchical machine learning system that predicts county-specific crop yield shocks and livestock forage deficits, aggregating data to provide macro-level regional intelligence and automated decision-support reporting.

Specific Objectives:

1. To identify key environmental, demographic, and historical indicators that predict crop failure and livestock forage deficits at the county level.
2. To build and compare the predictive performance of machine learning models (Logistic Regression as a baseline, versus Random Forest and XGBoost classifiers).
3. To integrate an Agentic AI (using NVIDIA NIM) that translates raw model outputs into plain-English insights, generates dynamic visualizations (maps/charts), and automates professional PDF reporting.
4. To deploy a functional, user-friendly Streamlit web application prototype for county agricultural officers and regional coordinators.

1.4 Research Questions

1. What are the most significant environmental and demographic predictors of crop yield shocks and livestock forage deficits in Kenyan counties?
2. How does the predictive performance (F1-Score, ROC-AUC) of advanced ensemble models (Random Forest, XGBoost) compare to a baseline Logistic Regression model for this task?
3. How effectively can an Agentic AI framework translate complex machine learning outputs into actionable, plain-English reports and dynamic visualizations for non-technical users?
4. What is the usability and practical value of the deployed Streamlit prototype compared to existing static agricultural dashboards?

1.5 Project Scope

The project focuses on predicting agricultural risk for **two primary domains**: staple crop production (e.g., Maize, Beans) and livestock forage availability. Geographically, the prototype will be trained and demonstrated on a representative sample of Kenyan counties (e.g., Uasin Gishu for crops, Kajiado for livestock), with the architecture designed to aggregate these predictions to a **Regional level** (e.g., Rift Valley, Eastern). The system will utilize open-source proxy data to build a fully functional prototype, deferring the integration of proprietary government databases to future work.

1.6 Limitations of the Study

1. **Data Availability:** Following capstone development best practices (Watanga, 2026), the initial prototype will be built using open-source proxy data and synthesized historical patterns to ensure rapid system development, deferring proprietary government data integration to the post-capstone phase.
2. **Computational Constraints:** Real-time Agentic AI inference via NVIDIA NIM APIs may be subject to rate limits or latency during the prototype demonstration phase.
3. **Scope of Variables:** The model will not account for sudden, unpredictable socio-economic shocks (e.g., sudden global fertilizer price spikes), focusing strictly on climate, environmental, and demographic-driven risks.

1.7 Significance of the Study (The 5 Critical Capstone Questions)

This project is designed to explicitly answer the five critical evaluation questions:

1. **What problem is being solved?** The reactive, uncoordinated, and siloed management of agricultural disasters (crop failure and livestock starvation) caused by a lack of interpretable, hierarchical early warning data.
2. **What is the developed solution?** AgriShield: A predictive, agentic AI decision-support system that forecasts county-level risks, aggregates them to a regional view, and auto-generated plain-English PDF reports.
3. **How was it built and tested?** Built via a rigorous ML pipeline (EDA → Feature Engineering → XGBoost) and an Agentic AI tool-calling architecture (NVIDIA NIM/LangChain). Tested by comparing model metrics (F1-Score, ROC-AUC) against a baseline, and evaluating the relevance of the Agent's generated reports.
4. **Who are the users?** County Agricultural/Livestock Officers, Regional Coordinators, the Ministry of Agriculture, and NGOs (e.g., FAO, Red Cross).
5. **How will it be sustained?** A scalable B2G (Business-to-Government) SaaS subscription model for the 47 counties, plus an API licensing model for agricultural insurance companies.

CHAPTER 2: LITERATURE REVIEW

2.1 Theoretical Literature

- **Climate-Smart Agriculture (CSA) Theory:** Emphasizes the need for predictive analytics to sustainably increase agricultural productivity and build resilience to climate change (FAO, 2023).
- **Predictive Analytics in Agriculture:** Theoretical frameworks supporting the use of historical and real-time data to forecast future agricultural outcomes, shifting management from reactive to proactive.
- **Agentic AI and Explainable AI (XAI) Theory:** Recent paradigms in computer science demonstrating that Large Language Models (LLMs), when equipped with "tool-calling" capabilities, can act as autonomous intermediaries to interpret black-box machine learning models and generate human-readable reports (Wang et al., 2024).

2.2 Empirical Literature

- **Early Warning Systems (EWS) in Sub-Saharan Africa:** Studies by Magrin et al. (2020) show that while satellite-based EWS exist, they frequently fail at the "last mile" due to overly technical outputs that local stakeholders cannot operationalize.
- **Machine Learning in Yield Prediction:** Research by Kamilaris & Prenafeta-Boldú (2018) confirms that ensemble tree-based models, particularly **XGBoost** and **Random Forest**, consistently outperform traditional linear models in capturing the non-linear, complex interactions between weather, soil, and crop yield.
- **LLMs in Agricultural Decision Support:** Recent work by Li et al. (2023) demonstrates that integrating LLMs with predictive models significantly improves user trust and adoption rates among non-technical farmers and government officials by providing natural language explanations.

2.3 Review of Existing Solutions & Competitor Analysis

To ensure AgriShield adds unique value and does not merely copy existing systems, we evaluated current market solutions. As highlighted in capstone mentorship guidelines, a competitor comparison is vital to justify the project's uniqueness.

Existing Solution Type	Examples	Pros	Cons / Flaws	How AgriShield Beats Them
1. Macro-level GIS Early Warning Systems	FEWS NET, FAO SWALIM	Highly accurate, uses satellite data.	Outputs complex GIS maps; requires a PhD to interpret; no automated reporting for local officers.	AgriShield uses an Agentic AI to translate complex GIS/ML outputs into plain-English text and auto-generated PDF reports.
2. Micro-level AgriTech Apps	Crop disease leaf scanners, basic weather apps	Easy for individual farmers to use via smartphones.	Strictly micro-level (single farm); ignores livestock; no county/region hierarchical view for government planning.	AgriShield operates at a County-to-Region hierarchy and integrates both crops and livestock forage data.
3. Standard ML Dashboards	Basic Streamlit yield predictors built by students	Good predictive accuracy; clean UI.	"Black box" outputs; static charts; user must manually interpret the data and write their own reports.	AgriShield features Dynamic AI Visualization (the AI draws the charts on the fly) and Automated PDF

Existing Solution Type	Examples	Pros	Cons / Flaws	How AgriShield Beats Them
				Reporting.

2.4 Research Gap

Based on the competitor analysis and empirical literature, two major gaps exist:

1. **The Hierarchical & Dual-Focus Gap:** Most AgriTech ML research focuses on micro-level problems (e.g., leaf disease), while macro-level models are complex global systems. There is a lack of **hierarchical (county-to-region) predictive models** tailored for East Africa that simultaneously handle **both crop yields and livestock forage**.
2. **The Interpretability & Automation Gap:** Existing predictive models output raw probabilities or static dashboards. There is a significant lack of systems that integrate **Agentic AI** to automatically translate these outputs into plain-English action plans, dynamic visualizations, and standardized PDF reports for government stakeholders.

This project directly addresses this gap by building AgriShield, combining hierarchical XGBoost prediction with NVIDIA NIM Agentic AI for automated, interpretable reporting.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

The project will employ a **Predictive Analysis and Experimental Research Design**. This involves training multiple machine learning algorithms on historical proxy data to predict future agricultural risk, and experimentally comparing their performance against a baseline model to determine the most accurate system.

3.2 Data Sources & Collection Strategy

To ensure rapid system development and avoid bureaucratic delays in acquiring proprietary government data, the project will adopt a **"Build-First" proxy data strategy**:

- **Meteorological Data:** Historical rainfall, temperature, and soil moisture will be fetched via

the **Open-Meteo API**.

- **Livestock/Forage Data:** NASA MODIS **NDVI (Normalized Difference Vegetation Index)** data will be used as a proxy for pasture greenness and forage availability.
- **Demographic & Crop Data:** **KNBS Open Data** and **FAOSTAT** will provide historical crop tonnage and county-level demographic profiles.
- *Note:* While the functional MVP will be built using these open proxies and synthesized historical patterns, the system architecture is designed to seamlessly ingest real-time Ministry of Agriculture APIs in the post-capstone deployment phase.

3.3 Data Science Pipeline

1. **Data Collection & Cleaning:** Automated Python scripts to fetch API data, handle missing values via imputation, and remove outliers.
2. **Exploratory Data Analysis (EDA):** Distribution analysis, correlation heatmaps, and time-series plots (e.g., rainfall anomalies vs. historical yield drops).
3. **Feature Engineering:** Creation of specialized agricultural KPIs, such as the "30-Day Rainfall Anomaly Index" for crops and the "Forage Dryness Index" for livestock.
4. **Model Development:** Training and hyperparameter tuning of predictive models.

3.4 Agentic AI Architecture (NVIDIA NIM Integration)

The system will utilize **LangChain** to orchestrate an Agentic AI workflow powered by **NVIDIA NIM** inference microservices:

- **Data Translator Agent:** Converts raw user inputs or messy CSVs into the strict JSON feature format required by the ML model.
- **Insight & Visualization Agent:** Uses "Tool Calling" to execute Python scripts (via Plotly and Folium) to generate dynamic, interactive maps and charts based on the ML model's output.
- **Reporting Agent:** Compiles the predictions, dynamic charts, and plain-English explanations into a standardized, branded PDF document using ReportLab.

3.5 Model Validation & Evaluation

The predictive models will be evaluated using standard classification metrics: **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**.

- **Baseline Comparison:** The complex model (XGBoost) will be strictly compared against a simple baseline model (Logistic Regression) to justify the computational cost and prove the complex model's superiority, as required by capstone guidelines.

3.6 Ethical Considerations

The project utilizes aggregate, open-source, and synthesized proxy data. No personally identifiable information (PII) or sensitive, individual farmer data is collected. This ensures full compliance with the **Kenya Data Protection Act (KDPA, 2019)**.

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